Lukas Minutello

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Master of Engineering, Mechanical Engineering — *Expected Dec 2026*
Bachelor of Science, Mechanical Engineering — *May 2026*

EXPERIENCE

ASML, Wilton, CT, *Mechanical Design Intern* — **May 2026 – Aug 2026**

- Aided development of the UNICOM for new iterations of EUV machines.
- Redesigned heat sink to be additively manufacturable (PBF) for optimized cooling channel geometry.
- Conducted temperature testing to verify FEA results and to quantify thermal performance gains.

Cornell Baja Racing, Ithaca, NY — **Oct 2022 – May 2026**

- Design, manufacture, and assemble an off-road vehicle to compete in the SAE Collegiate Baja Design Series.
- 50+ hours of manual mill and lathe experience.

Brakes Subteam Member — *Jul 2025 – May 2026*

- Designed floating brake rotors with reduced weight, minimized brake drag effects, less warping potential, more reliable pad contact, and better serviceability/ease of assembly.
- Conducted temperature testing on legacy rotors for raw temperature values and to quantify effects of brake drag.
- Performed coupled structural-thermal FEA in ANSYS using braking load cases and temperature logs to visualize stress and thermal expansion, informing material selection, air gap size, and other design parameters.

Suspension Subteam Member — *Jul 2023 – Jun 2025*

- Modeled the front uprights in SolidWorks to comply with tighter constraints prompted by the suspension geometry.
- Performed FEA in ANSYS using load cases derived from wheel force transducer testing, informing optimal locations for material addition/removal and retaining legacy FOS despite tighter packaging.
- Designed the tie rods and steering column using carbon fiber tubing and aluminum inserts for weight reduction.
- Developed rigorous adhesive bonding processes achieving a 50% increase in bond shear strength as measured by torsion and pull-out testing.
- Modeled carbon fiber tie rod tubes in ANSYS ACP using manufacturer layup data and performed Euler buckling analysis to validate the reported critical load.

Cornell Architectural Robotics Lab, Ithaca, NY, *Research Intern* — **May 2025 – Aug 2025**

- Developed CAD assembly for the *home+* robot and generated URDF files for motion planning in ROS2/RViz.
- Modified Arduino serial control code to send motion planning results to the *home+* robot.
- Cut and welded several aluminum L-channel square frames to be used as moving platforms for *Robot-Room.*

NYC Health + Hospitals/Woodhull, Brooklyn, NY, *Biomedical Equipment Technician Intern* — **Jul 2024 – Aug 2024**

- Troubleshot and resolved 50+ mechanical and electrical issues in multiple hospital departments.
- Performed planned maintenance procedures and electrical safety inspections on patient care equipment.
- Processed and closed 1,000+ work orders in the Crothall device database to ensure accurate equipment tracking.
- Optimized work order assignment and documentation to streamline the workflow of senior BMETs.

SKILLS

Design: CAD (NX, SolidWorks, Fusion 360), FEA (ANSYS, NX), DFMA, DFMEA, GD&T
Manufacturing: Machine Shop Trained (Lathe, Mill), CNC Operation and Programming, 3D Printing, Soldering
Programming: MATLAB, Python, LaTeX, Excel
Certifications: SolidWorks CSWA, Microsoft Office Specialist